ROBOTICS AND AUTOMATION COURSE

A nine week, hands-on course for grades 6 to 9 that ends with every student building a maze-solving Arduino robot they get to keep.

GRADES	LENGTH	SESSION	CAPSTONE
6 to 9	9 weeks, 1 session a week	2 to 2.5 hours	Maze-solving robot. Kits go home

This is a nine week, build-first robotics course. No technical background is needed. Each class is a short lesson followed by a long lab where students build, code, and troubleshoot. The first five weeks cover the fundamentals one piece at a time. The last four put them together into a robot that finds its own way through a maze, and students keep their kits when it's over.

It's designed for teams and partners who want to run something longer than a one-day event: an after school club, a library series, or a weekend program at a community center.

COURSE AT A GLANCE

WEEK	TOPIC OR PROJECT	SKILLS LEARNED
1	Arduino 101 and LEDs	Circuit basics, setting up the editor
2	Inputs and breadboards	Digital vs analog, wiring
3	Sensors	Reading inputs, if statements
4	Servos	Motor control, PWM
5	Integrated project	Combining inputs, outputs, and logic
6	Robot base build	Chassis, mounting, wiring safety
7	Bumpers and sensors	Prototyping, first drive tests
8	Navigation code	Autonomous programming
9	Capstone showcase	Testing, debugging, presenting

HOW EACH SESSION RUNS

PART	TIME	WHAT HAPPENS
Concept of the day	20 to 30 min	Short talk and live demo of the new component or idea
Lab time	60 to 75 min	Students build and code with instructors circulating
Break	15 min	Snack and movement, away from the benches
Project work	30 to 45 min	Troubleshoot, improve, and save working code

WEEKLY PLAN

WEEK 1 | INTRODUCTION AND ARDUINO BASICS	
TOPICS	Lab safety, the course plan, a tour of the Arduino board and its pins, what each kit part does, setting up the editor.
LAB	Breadboard practice, powering an LED, uploading and editing the Blink example.
OUTCOME	Every student powers and programs their first circuit.
PREP	Test every laptop type in advance. Pre-sort kits so each student gets one bag.

WEEK 2 | DIGITAL VS ANALOG INPUTS	
TOPICS	Digital and analog signals, series vs parallel circuits, reading a resistor.
LAB	Read a push button. Read a potentiometer and print the values to the Serial Monitor.
OUTCOME	Modify Blink to react to a button, then make an LED that dims with a knob.
PREP	Have a multimeter on hand to show voltage changing as the knob turns.

WEEK 3 | SENSORS AND CONDITIONALS	
TOPICS	What sensors do, using if statements to react to sensor readings.
LAB	Test limit switches and a color sensor. Print readings and set thresholds.
OUTCOME	An LED or buzzer turns on when a sensor changes state.
PREP	Bring a few colored cards for the color sensor so readings are consistent.

WEEK 4 | SERVO MOTORS AND OUTPUT CONTROL	
TOPICS	How servos work, pulse width modulation (PWM), positional vs continuous servos.
LAB	Control a servo's position with a button or a potentiometer.
OUTCOME	Each student completes a simple moving part.
PREP	Show both servo types side by side. Students mix them up later if they don't see the difference now.

WEEK 5 | COMBINING INPUTS AND OUTPUTS	
TOPICS	Using sensors and motors in the same program, a step-by-step way to troubleshoot.
LAB	Build a security gate: a servo arm lifts when a sensor is triggered, with a buzzer alarm.
OUTCOME	Students demo a working system and explain how they debugged it.
PREP	This is the midpoint. Check in with students who are behind before the capstone starts.

WEEKS 6 AND 7 | CAPSTONE PLANNING AND CHASSIS	
TOPICS	What maze solving is, bumper logic, a simple design process: plan, build, test, change.
LAB	Plan the build in steps, assemble the robot base, mount the Arduino and battery, wire safely.
OUTCOME	The robot drives with simple code and the bumpers are tested.
PREP	Set up the practice maze by the end of week 6 so students can test against real walls.

WEEKS 8 AND 9 | PROGRAMMING, TESTING, AND SHOWCASE	
TOPICS	Autonomous navigation, loop strategies, tuning turn times, cleaning up code.
LAB	Fine-tune navigation, run timed maze trials, then the final showcase in week 9.
OUTCOME	A finished maze-solving robot. All kits go home with students.
PREP	Invite families to week 9. Print certificates ahead of time.

THE CAPSTONE ROBOT

The robot drives on two continuous rotation servos and uses two limit switches on the front as bumpers. When a bumper hits a wall, the robot backs up, turns away, and tries again. It's a simple strategy, but it gets through most beginner mazes, and students can improve it in a dozen ways.

STARTER CAPSTONE CODE

```
// Bump-and-turn maze robot
#include <Servo.h>

Servo leftWheel, rightWheel;
const int LEFT_BUMP = 2;
const int RIGHT_BUMP = 3;

void drive(int l, int r) { leftWheel.write(l); rightWheel.write(r); }
void forward()   { drive(180, 0); }    // wheels face opposite ways
void reverse()   { drive(0, 180); }
void turnLeft()  { drive(0, 0); }
void turnRight() { drive(180, 180); }

void setup() {
  leftWheel.attach(9);
  rightWheel.attach(10);
  pinMode(LEFT_BUMP, INPUT_PULLUP);
  pinMode(RIGHT_BUMP, INPUT_PULLUP);
}

void loop() {
  if (digitalRead(LEFT_BUMP) == LOW) {        // hit on the left
    reverse();   delay(400);
    turnRight(); delay(350);
  } else if (digitalRead(RIGHT_BUMP) == LOW) { // hit on the right
    reverse();   delay(400);
    turnLeft();  delay(350);
  } else {
    forward();
  }
}
```

Turn times depend on the battery, the floor, and the wheels. Have students measure a 90 degree turn on their own robot and change the 350 to match.

SUGGESTED CAPSTONE RUBRIC

AREA	POINTS	WHAT FULL POINTS LOOKS LIKE
Maze navigation	40	Robot reaches the end of the maze in at least one of three runs
Code reliability	30	Code is readable, commented, and works the same way run after run
Documentation	20	A short build log: what they tried, what broke, how they fixed it
Showcase	10	Explains how the robot decides what to do

MATERIALS

IN EACH STUDENT KIT

- Arduino Uno or compatible (not a Mega)
- Breadboard and jumper wire set
- Resistors and LEDs
- Positional servo and 2 continuous rotation servos
- 2 limit switches and a color sensor
- Robot chassis with wheels
- USB cable

INSTRUCTOR SUPPLIES

- Maze walls (cardboard or foam board works)
- Tape, zip ties, and spare parts
- A few extra charged battery packs
- Projector for demos

STUDENTS BRING

- A laptop with Google Chrome
- School Chromebooks are fine
- Windows or Mac laptops are fine
- A snack and a water bottle

STAFFING AND TIPS

Staff: 1 lead instructor and 1 assistant for every 10 students. An FTC team can easily cover this with 3 or 4 members rotating weeks.

- Debug in small steps. Check wiring and code after every change so problems are easy to find.
- Save every working version. Have students save a copy with the week number before changing anything (maze_week8_v2).
- Show, don't just tell. Pass new components around and project a close-up before students wire them.
- Pair students thoughtfully. Put a more confident student with one who's newer. Both learn more.
- Keep a parts bin. Something will break every week. A spare servo saves a whole lesson.

EXTENSIONS FOR STUDENTS WHO FINISH EARLY

- Line following with an IR sensor instead of bumpers.
- Wall following: keep one bumper lightly touching the wall to trace the maze.
- Add an ultrasonic distance sensor so the robot turns before it hits anything.

QUESTIONS? TALK TO A HUMAN.

Email sparkgearsinitiative@gmail.com or call 571-429-9220. Request a mentor and a supply kit at sparkgearsinitiative.org/for-teams.html. We usually reply within two school days.